Fire death atlas: observed cessation of detectable spatial growth

Informal title note: 'fire death' is used here only as shorthand for observed cessation of detectable spatial growth in cached FIRED non-prescribed candidate fires.

Candidate fires: 25; fire-time records: 229; primary QC pass: 1.

Final area range: 1.29-261.24 km2.

Observation count range: 5-14 records per fire.

Available climate variable used: gridMET daily maximum temperature (tmmx), converted from Kelvin to Celsius.

Primary limitation: climate is nearest grid cell to event centroid. Newly burned area, cumulative footprint, and advancing-boundary climate are not separately estimated because these candidate fires are often smaller than gridMET cell support.

No prescribed-fire comparisons are made or implied.

Data availability

requested_variable	available	units	temporal_resolution	spatial_support	used_in_analysis
fire polygons/perimeters	True	FIRED sinusoidal projected CRS, metre units	irregular daily detections	FIRED candidate event polygons	True
fire IDs	True	integer id	event-level/time-level	FIRED id	True
timestamps	True	calendar date	detected timesteps	daily records	True
maximum temperature	True	deg C after K conversion	daily	nearest grid cell to event centroid	True
minimum temperature	False	unavailable	daily requested	gridMET	False
vapor pressure deficit	False	unavailable	daily requested	gridMET	False
relative humidity	False	unavailable	daily requested	gridMET	False
wind speed	False	unavailable	daily requested	gridMET	False
precipitation	False	unavailable	daily requested	gridMET	False
fuel moisture	False	unavailable	daily requested	gridMET	False
drought index	False	unavailable	daily requested	gridMET	False
elevation	False	unavailable	static	context	False

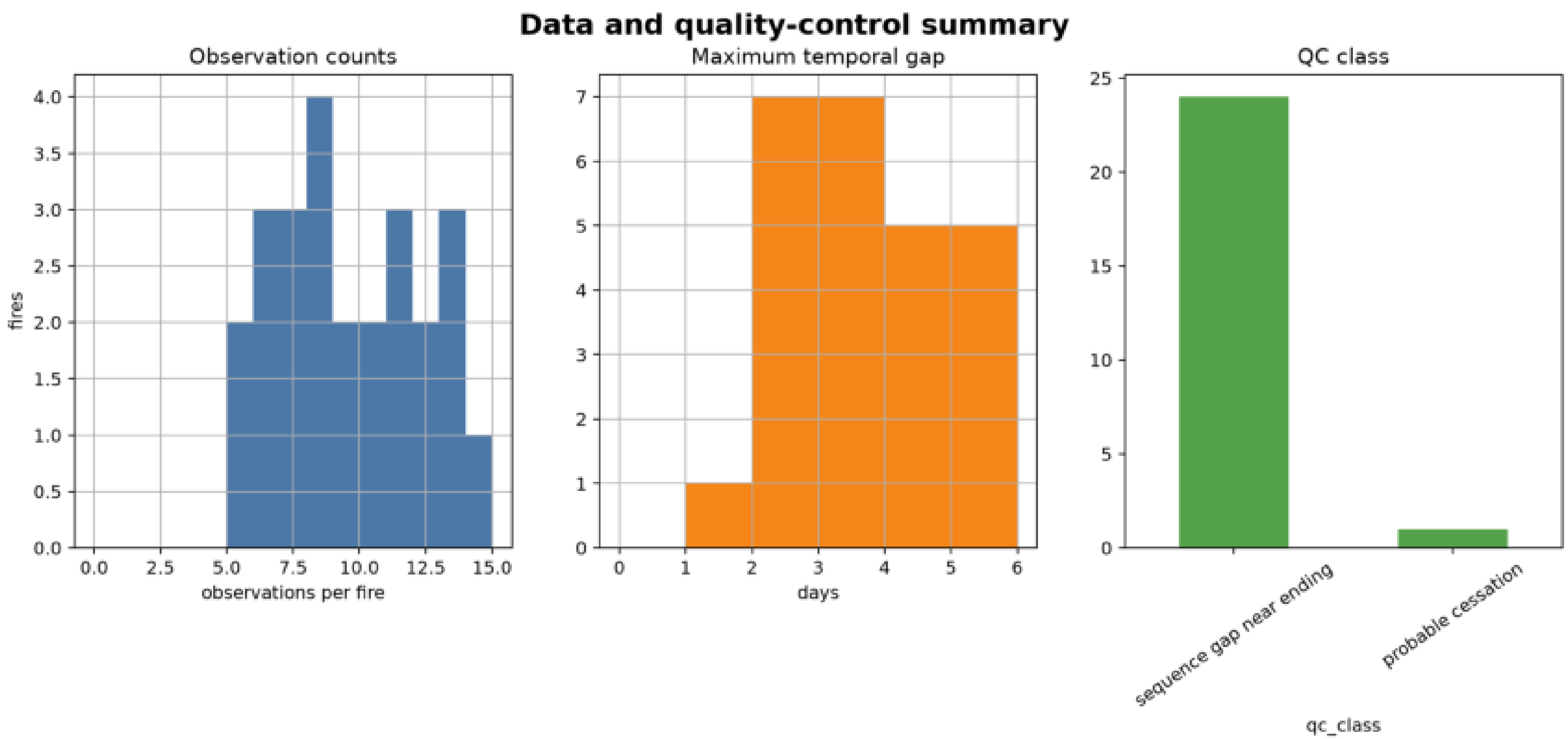
Summary table

metric	n	min	median	mean	max
final_area_km2	25	1.29	7.08	31.9	261
duration_days	25	14	14	14	14
observation_count	25	5	9	9.16	14
peak_newly_added_area_km2	25	0.429	2.79	10.3	65.5
terminal_tmmx_c	25	11.4	24.2	23.6	36
terminal_growth_km2	25	0.215	0.215	0.412	2.58
fires_total	25	nan	nan	nan	nan
fires_primary_quality_pass	1	nan	nan	nan	nan
fire_time_rows	229	nan	nan	nan	nan

Quality-control sample-size flow

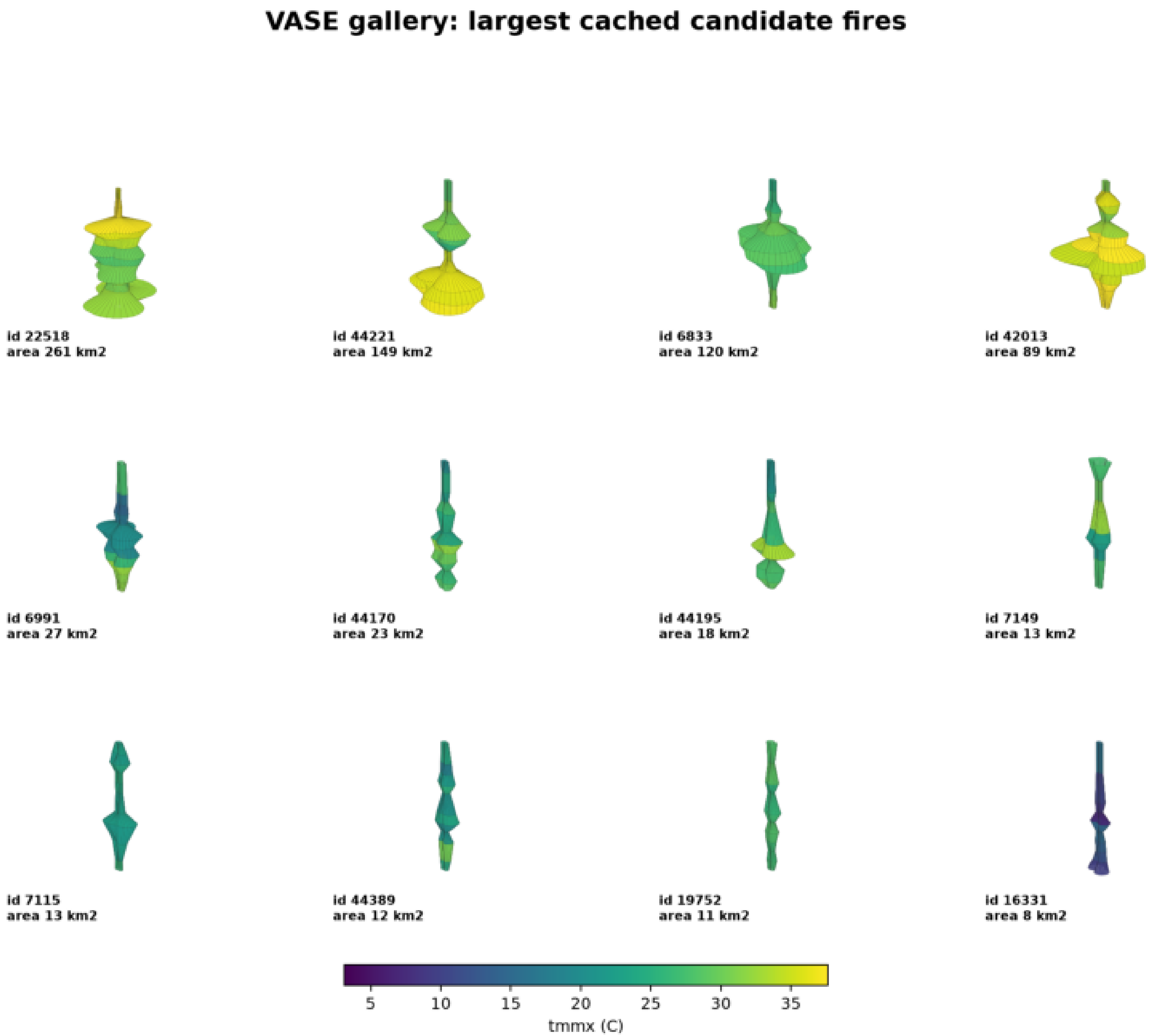
fire_id	observation_count	event_duration_days	max_gap_days	qc_class	primary_quality_pass
2445	8	14	5	sequence gap near ending	False
6833	13	14	2	sequence gap near ending	False
6991	12	14	3	sequence gap near ending	False
7115	11	14	2	sequence gap near ending	False
7149	9	14	3	sequence gap near ending	False
7421	8	14	3	sequence gap near ending	False
7508	8	14	3	sequence gap near ending	False
16331	10	14	3	sequence gap near ending	False
17532	7	14	4	sequence gap near ending	False
17903	5	14	4	sequence gap near ending	False
18606	7	14	4	sequence gap near ending	False
19752	11	14	2	sequence gap near ending	False
22518	14	14	1	probable cessation	True
22615	6	14	5	sequence gap near ending	False
23942	8	14	4	sequence gap near ending	False
24184	7	14	3	sequence gap near ending	False

Data and quality control



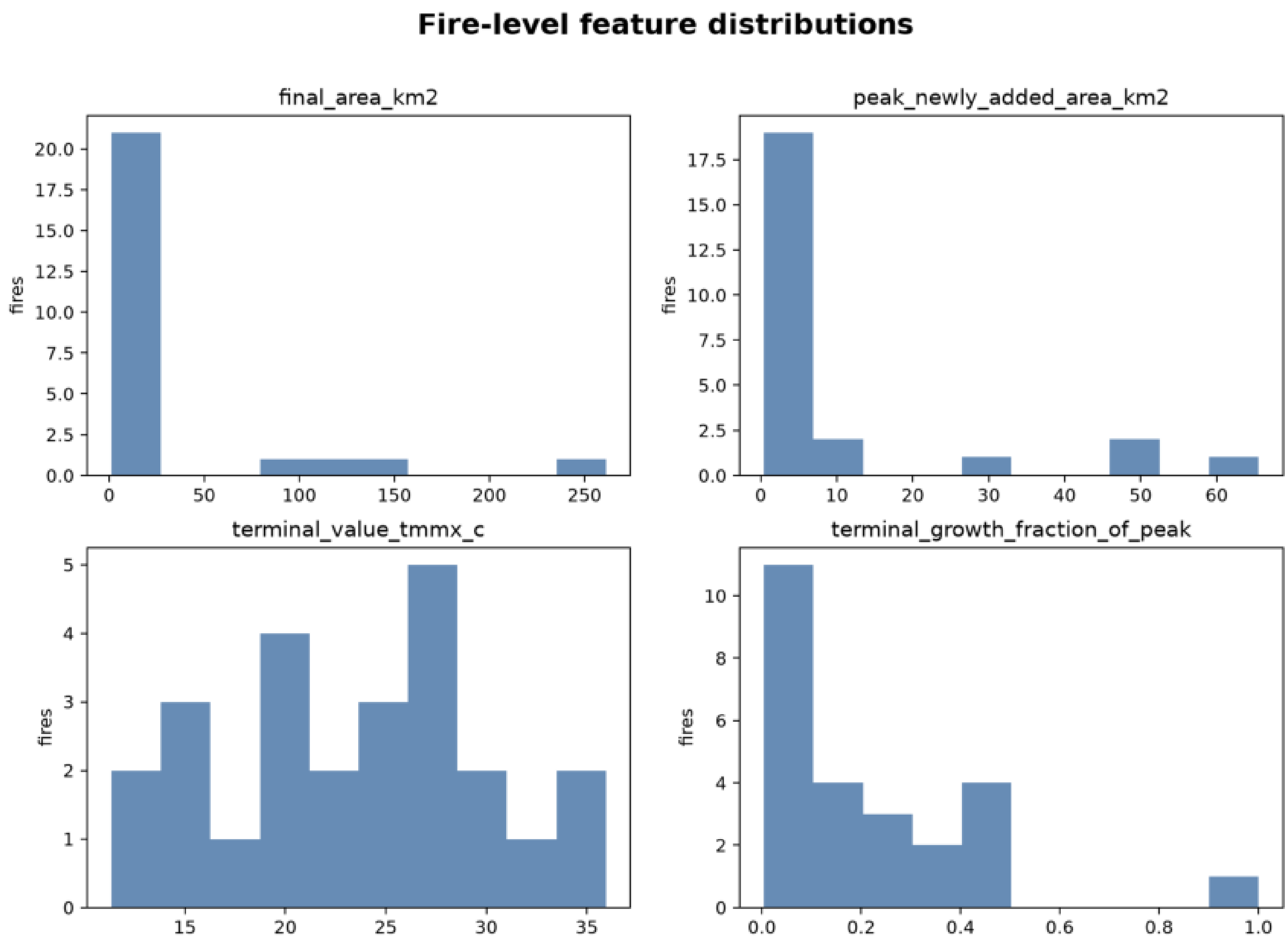
Most candidate endings have sequence gaps; interpret terminal observations as dataset-observed endings, not physical extinction.

VASE gallery with common tmmx scale

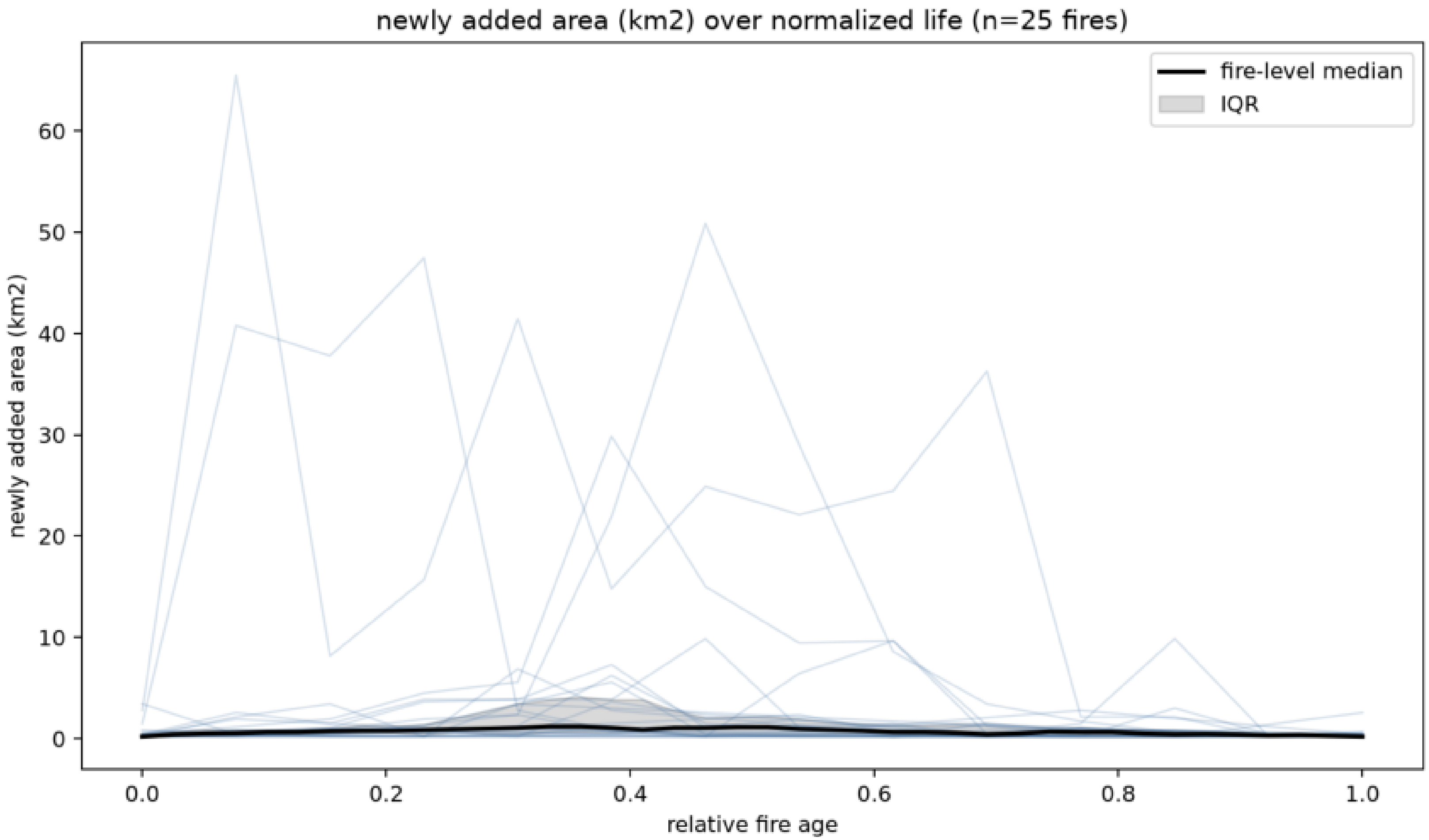


Gallery uses common tmmx color scale and largest fires by final area.

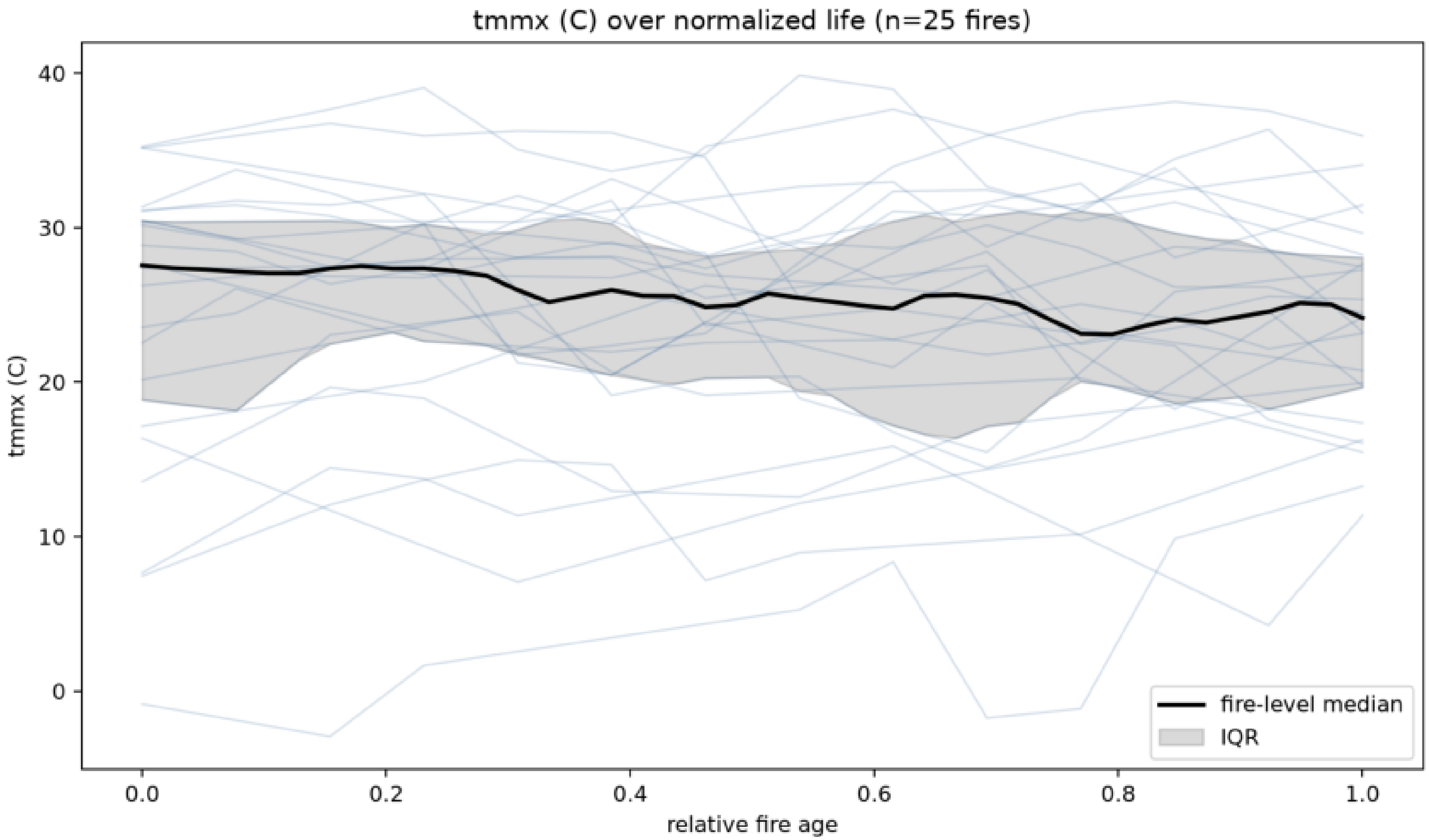
Fire morphology distributions



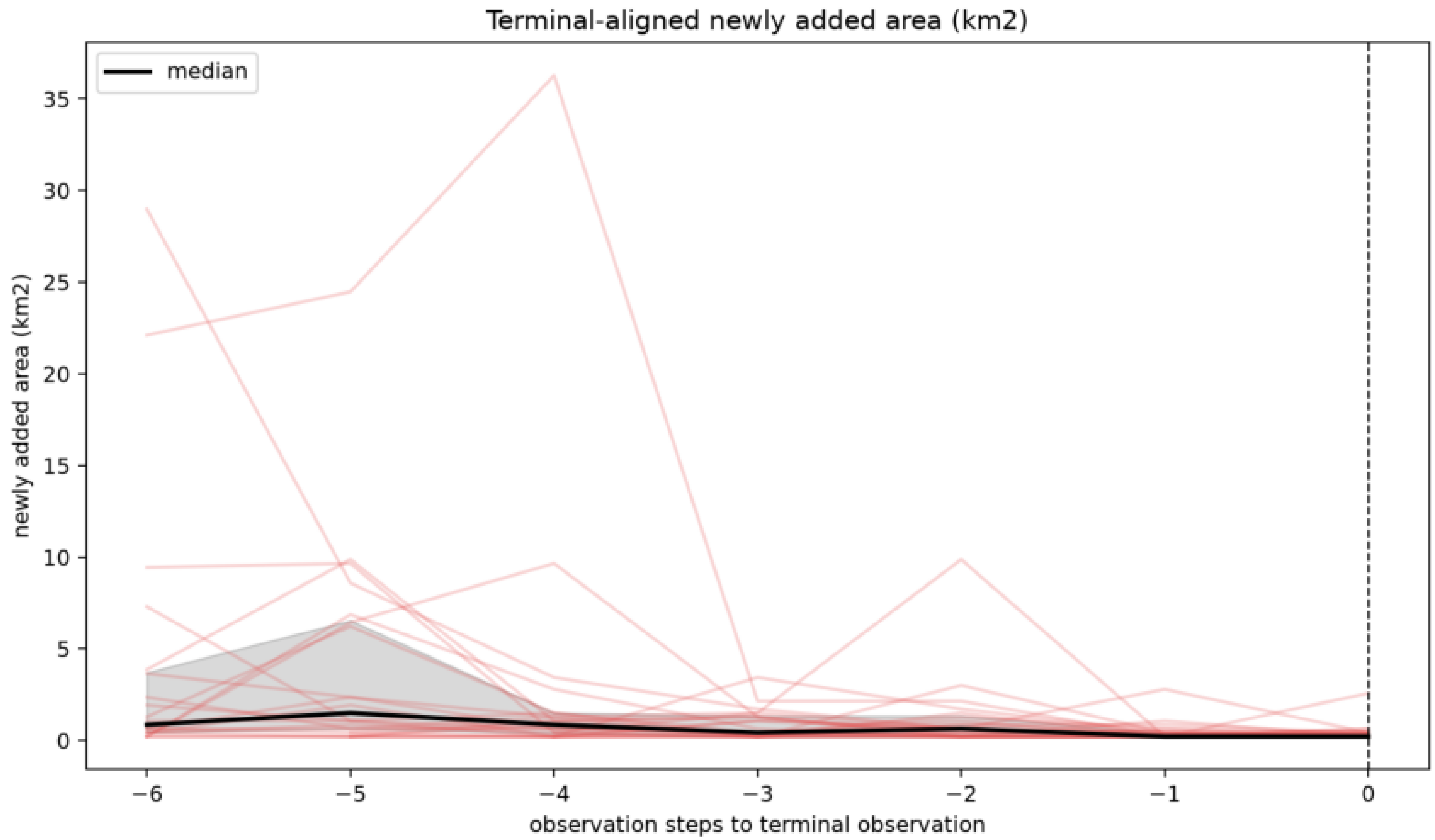
Complete life histories: growth



Complete life histories: maximum temperature

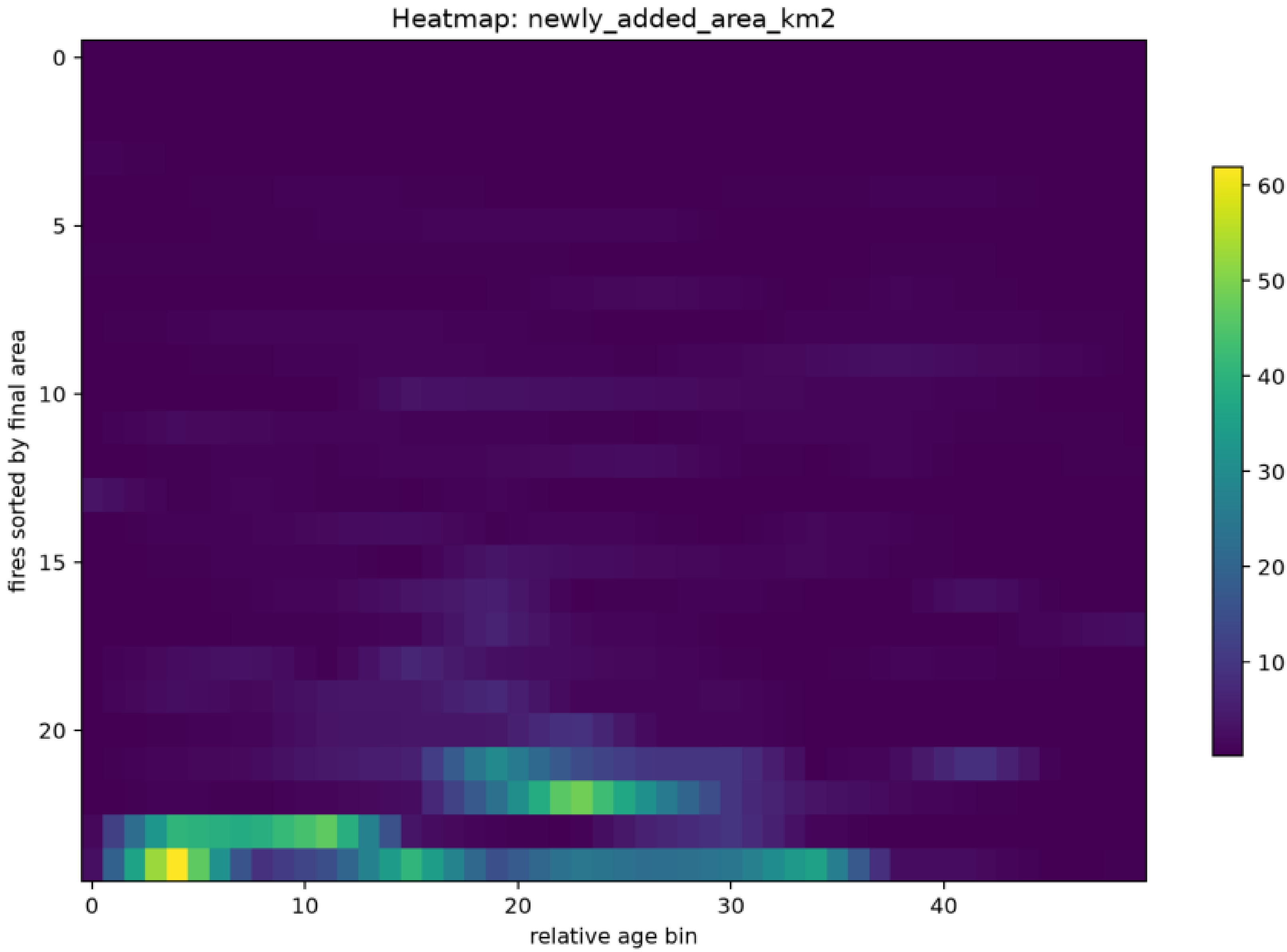


Terminal alignment: geometric activity

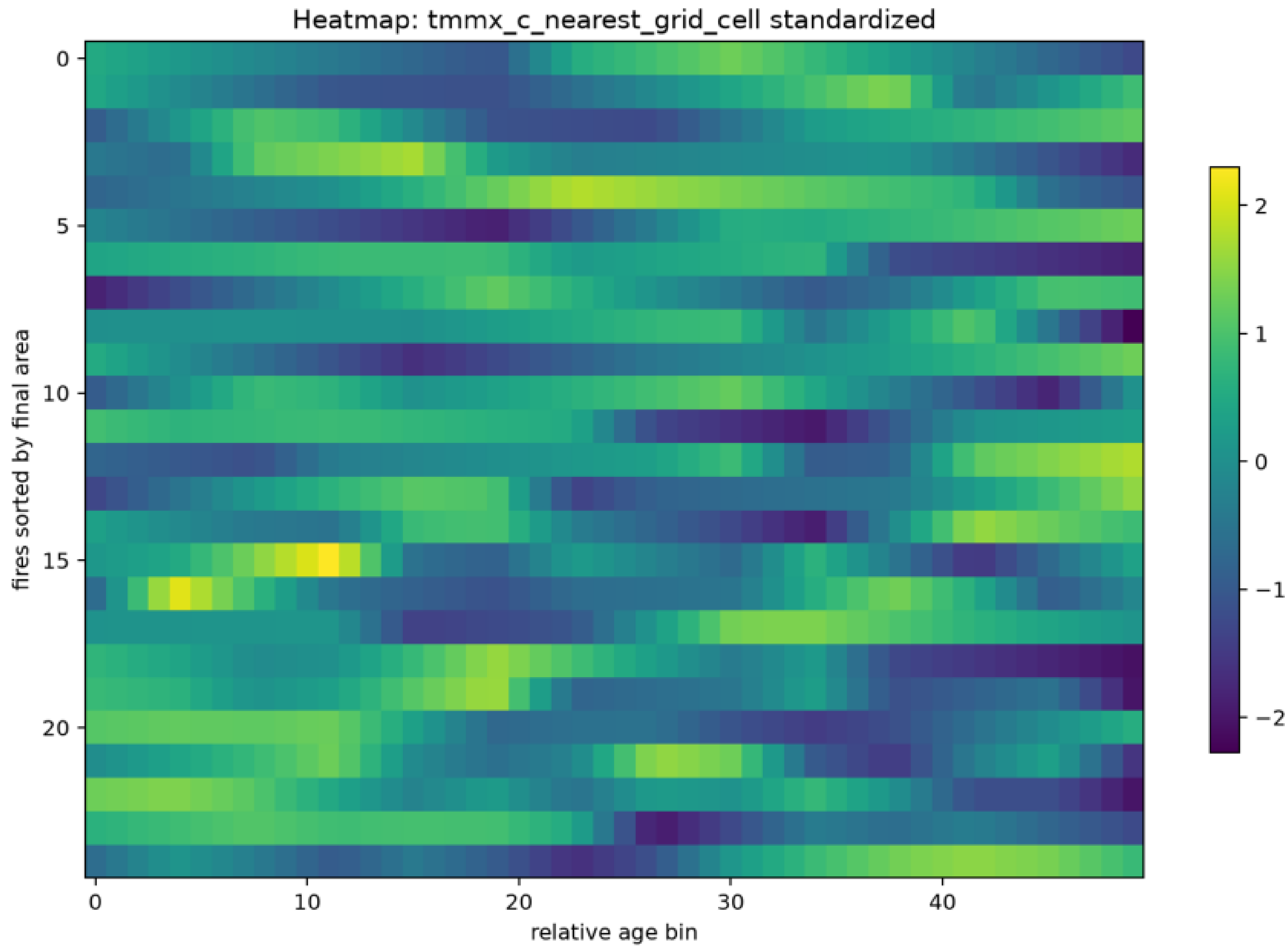


Terminal alignment: within-fire tmmx anomaly

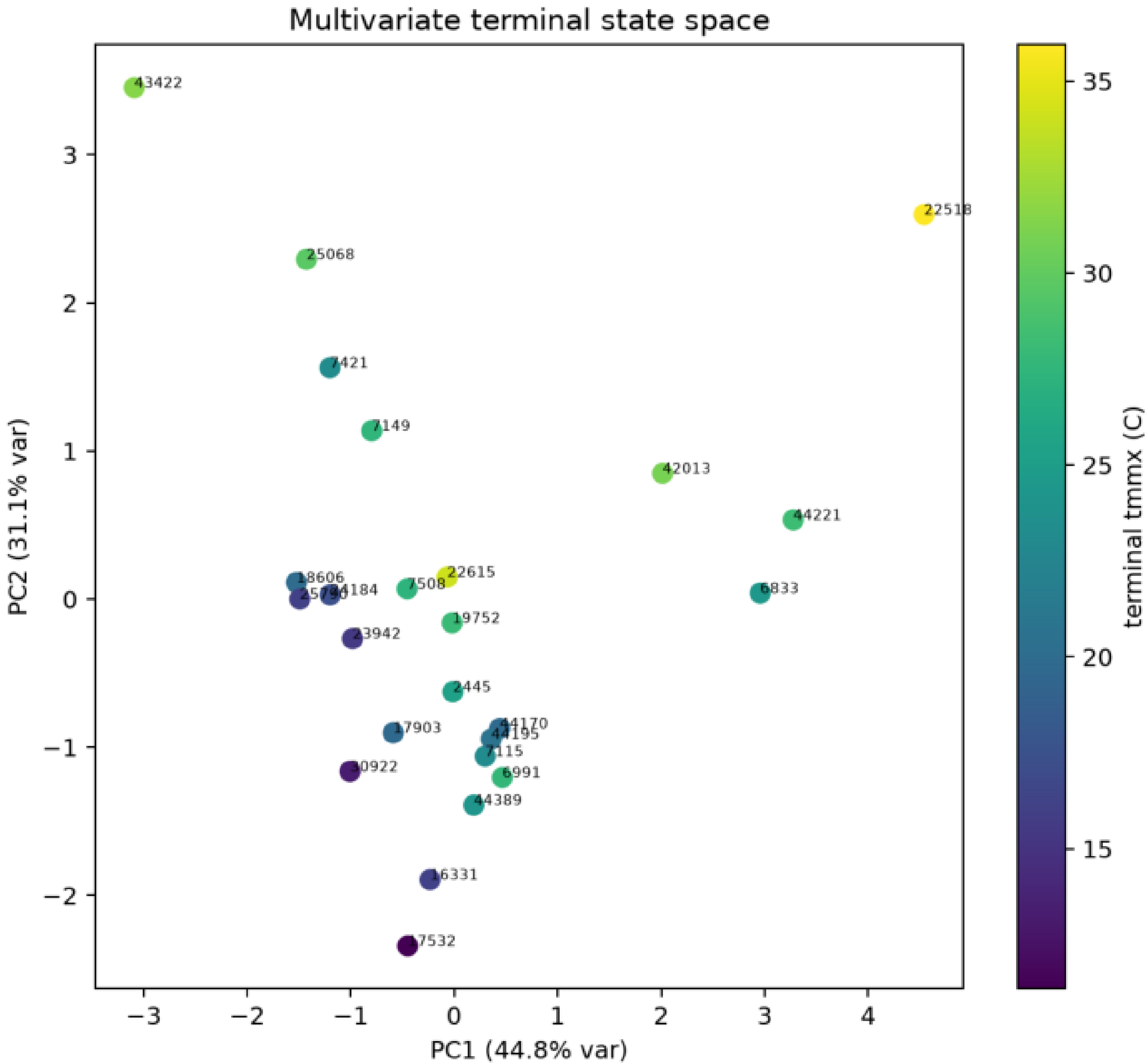
Heatmap: growth over relative age



Heatmap: standardized temperature over relative age



Multivariate terminal state space



Candidate scientific decisions

A. Common absolute terminal state: weak evidence with current data; terminal tmmx remains geographically/seasonally heterogeneous.

B. Within-fire anomaly convergence: ambiguous; standardized terminal trajectories do not justify a strong convergence claim.

C. Common direction without common endpoint: plausible for geometric growth decline by construction, but climate direction is mixed.

D. Multiple recurring terminal pathways: possible, but sample size is too small for stable clustering.

E. Climate alone explains cessation: not supported from this exploratory atlas; geometry and observation process dominate the current evidence.

F. Data quality/censoring limits terminal-process claims: strong caution. Many fires have irregular observation gaps near the observed ending.

Recommended next analysis: add VPD, wind, precipitation, fuel moisture, and climate normals; expand beyond the 25 candidate fires; compute climate on newly burned and advancing-boundary supports at a spatial resolution matched to the fire perimeters.